

### Simulation engine (adversary\_ladder.py)

8-direction votes, vector mean · fixed speed 5 m/s · delay 433 ms  
look-ahead 2 m · 65 s horizon · zigzag: periodic path (corrected)

### Published grid (T0 / T1 / T2)

4 geometries × 10 tr × 12 N × 3 adversary models  
MC = 50: 72,000 runs (zigzag re-simulated: 23,400)  
raw-run ANOVA 3,600 · speed 3,600  
behaviour panel 12,000 · lag axis 800 · mechanism 400

### Revision campaigns

geometry scale sweep 27,000 · radius extension 3,600  
polygon shape sweep 25,200 · large crowds to 3,200: 8,000  
measured latency  $\tau = 867$  ms: 1,600  
vote-noise floor · corner response · localisation (~1,000)

### Analysis

ceiling fit  $a + b/\sqrt{N}$  (OLS, WLS)  
change-point in  $b(\text{tr})$   
two-way ANOVA  
floor law  $a(R) = a_0 + c/R$   
quadrature  $e^2 = e_{\text{lag}}^2 + e_{\text{adv}}^2$   
saturation onset,  $N_c(\delta)$

### Main text

Fig 1 paths  
Fig 2  $b(\text{tr})$ ,  $R^2$  map  
Fig 3 RMSE-N to 3,200  
Fig 4 adversary ladder  
Tables 1–4

### Supplement and archive

Tables S1–S9  
Figs S1–S5  
Zenodo: code, data  
(21 files), figure scripts